

CSU IS Career Hub

Final Production Release | Capstone 465

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The Centralization Gap

- ✕ CSU IS students struggle to find localized internship data.
- ✕ Salary expectations for entry-level roles are often opaque.
- ✕ Existing platforms lack alumni feedback specific to the major.
- ✕ Manual tracking of certification paths (CompTIA, AWS) is inefficient.



Engineering the Hub



Smart Filtering

Location, role, and experience-based job discovery.



Salary Insights

Real-world entry-level salary reports for IS roles.



Cert Pathways

Guided recommendations for AWS, Security+, and Azure.

A unified ecosystem connecting students, jobs, and data.

The Production Stack

Frontend & Design

-  React 18 & TypeScript
-  Tailwind CSS & Shadcn UI
-  Vite Build Tooling

Backend & Infra

-  Django 5.0 (REST Framework)
-  PostgreSQL (Production)
-  Render Hosting

Performance Impact

97

Lighthouse Performance

Measurable Improvements

Optimization: Implemented React Lazy Loading & Suspense.

Tradeoff: Brief 500ms initial spinner vs. 496KB payload reduction.

Result: Baseline 58 score → 97 Production score.

Hardened Standards



OWASP Framework

ASVS and WSTG mapped risk register for auth and validation.



WCAG 2.1

Accessible UI reaching consistent 93+ scores via WebAIM checklist.



Event Analytics

Custom JOB_VIEW logger for student interaction tracking.

Automated Delivery



GitHub Actions CI



Auto Build / Test



Render Deploy

Dual-environment workflow ensures backend logic checks (Python 3.12) and frontend compilation (Node 20) pass on every merge to main.

Sprint Advancement



Project Retrospective

What Went Well

Early CI/CD integration reduced bug triage time.

Shadcn UI ensured cohesive design system.

Lessons Learned

Accessibility must be audited early in Sprint 1.

Lazy loading is essential for modern React.

Beyond MVP



CSU SSO

Microsoft 365 integration for CSU authentication.



Resume Parsing

Automated skill matching using ML algorithms.



Recruiter Portal

Direct job management for local hiring managers.

Handoff Summary

Documentation: Full system guide in `/docs/handoff_package.md`

Credentials: Admin access secured via environment variables; initial keys provided.

Maintenance: Automatic dependency updates via Dependabot scheduled.

Known Issues: Easy Apply UI is mocked; S3 storage required for file uploads.

Questions?

Thank you for your time.

 github.com/Reuvi571/csu-is-career-hub
